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Advanced NLP

→ Statistical $\Rightarrow$ Attention $\Rightarrow$ Transformers
(Google Translate) Drawback: Latency $\rightarrow$ Parallelized
($\therefore$ RNN, LSTM)

(Arrival movie)

→ Creole } $\rightarrow$ Contact / Conflict
Pidgin }

Dialect $\rightarrow$ Isolation

A language is a dialect with Army and Navy.

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→ Word Representations:

One-Hot Encoding

$\Downarrow$

LLM Embeddings

- Backpropagation
- Anaphora — Pronouns
 - Co-Reference - Multiple Pronouns
 - Causal — Cause-Effect Relation
 - Temporal - Time-Based / Chronology
- Long Distance Relationships
- LSTM
 - $\hookrightarrow$ Transfer Learning not possible
- Attention
 - $\hookrightarrow$ No Explicit Alignment

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→ Self-Attention:

$$\max(f(q_i, k_j))$$

$$\hookrightarrow \text{MLP}_2(q_i, k_j; w_1, w_2)$$

$$\text{PE}(i) \rightarrow q_i$$

$$\text{PE}(j) \rightarrow k_j$$

Modified $\leftrightarrow$ Modifier relationship - Relevance

Semantic Relationship - No Grammatical relevance

MLP₂ → Computationally expensive, but Complete

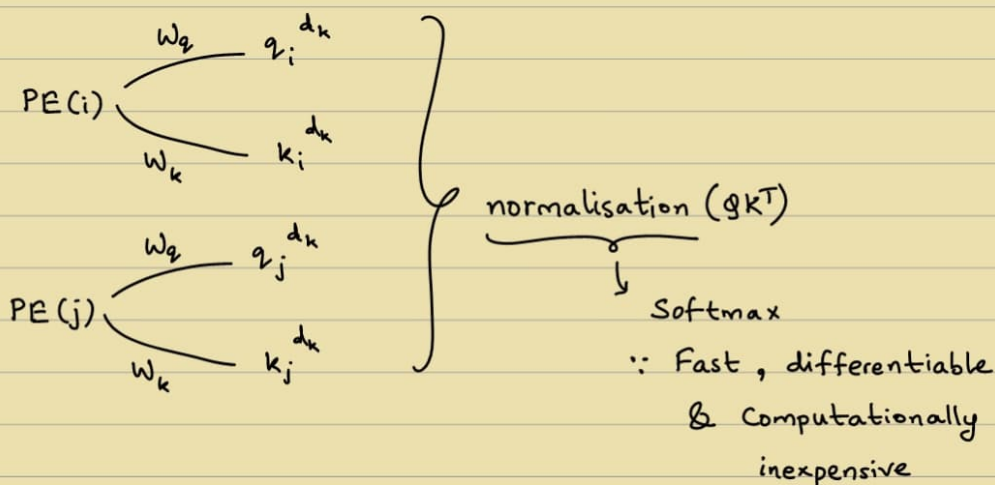
∴ $q_i \cdot k_j$ → Matrix Multiplication
 ϕk^T → Faster by GPUs

$q_i W k_j$
↓
Complete
but not fast

Cosine similarity → $\frac{q_i \cdot k_j}{\|q_i\| \cdot \|k_j\|}$

Linear Transformation → Projection

PE(i) - LT ,
PE(j) - LT ,
d dimension → Project it to d_k
i.e. 1/8th ∴ 8 heads



$$\text{Softmax} \rightarrow \frac{e^{\text{score}_{ij}}}{\sum_j e^{\text{score}_{ij}}}$$

$\sqrt{d_k} \rightarrow$ Dimensionality constant

V carries meaning
Computationally efficient

$c_i = \sum \text{attention}_{(i,j)} \cdot V$
→ context representation for the word i

- $o_i = \sigma(\sum_i w_i h_i + b)$
 $h_i = \sigma(\sum_i w_i c_i + b)$
- Neural Networks : Projects / Maps into another space / dimension
- VC dimension

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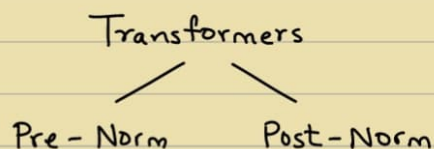
→ Research $\Rightarrow$ Design $\Rightarrow$ Engineering

↑↑

Abstraction and Understanding (Theory)

→ Self Attention

Multi-Head Attention

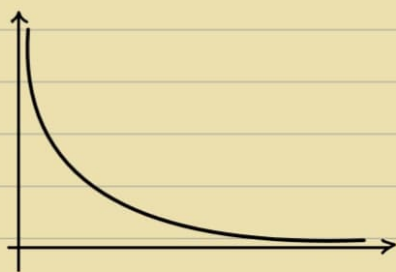


→ Morpheme

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Token: Identifier for a piece of information

Zipf's Curve

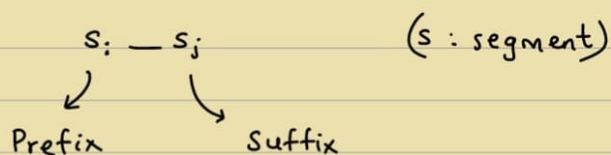


3B Words

↓ 4 byte space

12 GB models

10M × 10M → Sparse Matrix
∴ d × 10M



$$\begin{aligned} P(s_i, s_j) &= P(s_j | s_i) \cdot P(s_i) = P(s_i | s_j) \cdot P(s_j) \\ &= \frac{P(s_i \cap s_j)}{P(s_i) \cdot P(s_j)} \end{aligned}$$

- WordPiece

SentencePiece

Byte-Pair Encoding

3 % of Eng. vocab accounts for 95 % of the Eng. tokens

{ Byte → Byte-Pair → Character → Subword → Word }

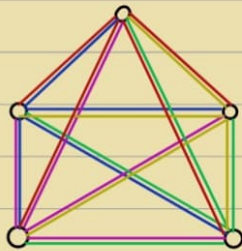
No tokenisation $\rightarrow$ e.g. MegaByte

$\hookrightarrow$ Patches of 4 bytes as Token

$\therefore$ Byte Latent Transformer

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$$c_i = \sum_j \text{att}(i,j) (E_j)$$



Every word pays attention to every other word

Ordering of the words within a language.
(Relative Ordering)

- Absolute PE
- Sinusoidal PE
 - Static
 - $\sin()$, $\cos()$ $\rightarrow$ Ratio

Periodic $\Rightarrow$ Phase difference

$[-1, +1]$ $\Rightarrow$ Confined

$$\therefore \begin{aligned} &\sin(\text{pos}/10000^{2i/d}) \\ &\cos(\text{pos}/10000^{2i/d}) \end{aligned}$$

- Relative PE
- Rotary PE (RoPE) - 2021
- NoPE
 - $\hookrightarrow$ No PE - Causal Mask Approach
 - $\hookrightarrow$ Not preferred

* PE $\rightarrow$ Location

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- KV Cache
- Multi Head Latent Attention
- Log Linear Attention

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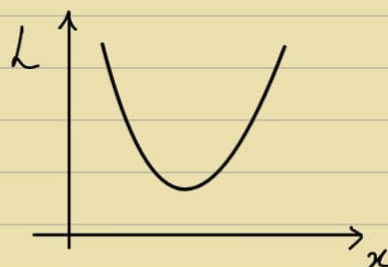
→ Normalization:

$y = ax_1 + bx_2$, Linear Regression

$$L = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i(a, b))^2 \leftarrow \text{MSE}$$

Convex loss function

$$x_i = \frac{x_i - E[x_i]}{\text{Var}[x_i]}$$



$6\sigma \rightarrow 99.9999\%$
(Gaussian)

Normalization

- Reduce Overfitting
- Model Generalises better
- Speed of Convergence $\uparrow$
- Sensitivity to input feat. $\downarrow$

• Batch Norm :

→ index-wise

Disadv. : Unknown Batch Size
Can be 1

• Layer Norm

• RMS Norm

→ Layer Norm with $\mu, \epsilon = 0$

• Weight Norm (x) :

$$\frac{g}{\|v\|} v^T x + b$$

→ On the parameters

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→ Residual Connections :

$$x_{L+1} = \sigma(x_L + F(x_L)) \leftarrow \text{Mitigates Vanishing Gradients}$$

Post Norm :

$$\text{Post} = \text{LN}(x + F(x))$$

Training Unstable

No loss of information

Pre Norm :

$$\text{Pre} = x + F(\text{LN}(x))$$

Avoids Vanishing Gradient

Representations Collapse

Deep Norm : PostNorm Scaled

$$x_{L+1} = \text{LayerNorm}(\alpha x_L + F(x_L))$$

Hybrid Norm

- Pre-Norm : Small Networks

Deep-Norm : Larger Networks

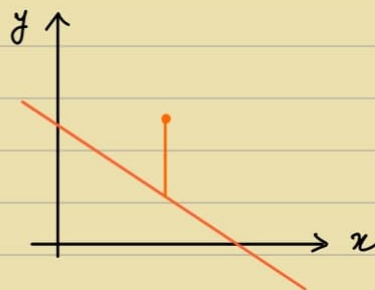
- FFNN → Non-Linearity

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(VC Dimensions)

$$y = \left(\sum w_i x_i + b \right)$$

$$\hat{w} = w + \eta \cdot \frac{\partial E}{\partial w}$$



• ReLU

• Gating (GLU)

↳ Efficiency of Training ↑

Bilinear

★ Kimi - K3 (Sigmoid tanh) - Read Paper
↳ Larger Model

→ Scaling : MoE

Expert Router